

Assignment 1

Part 2 · Rules of Probability

Worked solutions to Questions 1–6

Question 1 · Independent events

Given $P(A) = 0.4$ and $P(B) = 0.3$, and A and B are independent:

(a) By independence,

$$P(A \cap B) = P(A)P(B) = 0.4 \times 0.3 = \mathbf{0.12}.$$

(b) By the addition rule,

$$\begin{aligned} P(A \text{ or } B) &= P(A) + P(B) - P(A \cap B) \\ &= 0.4 + 0.3 - 0.12 = \mathbf{0.58}. \end{aligned}$$

Question 2 · Testing independence

Given $P(A) = 0.5$, $P(B) = 0.4$, and $P(A | B) = 0.7$. Since $P(B) > 0$, independence would require $P(A | B) = P(A)$. But $0.7 \neq 0.5$, so **A and B are not independent**.

Equivalently, $P(A \cap B) = P(A | B)P(B) = 0.7 \times 0.4 = 0.28$,
whereas $P(A)P(B) = 0.5 \times 0.4 = 0.20$. These are unequal.

Question 3 · Bayes' rule

Given $P(A) = 0.6$, $P(B | A) = 0.5$, and $P(B | A') = 0.2$.

First, $P(A') = 1 - 0.6 = 0.4$.

By the law of total probability,

$$\begin{aligned} P(B) &= P(B | A)P(A) + P(B | A')P(A') \\ &= 0.5 \times 0.6 + 0.2 \times 0.4 = 0.30 + 0.08 = 0.38. \end{aligned}$$

Then Bayes' rule gives

$$\begin{aligned} P(A | B) &= P(B | A)P(A) / P(B) \\ &= 0.30 / 0.38 = 15/19 \approx \mathbf{0.789474 \text{ (78.9474\%)}}. \end{aligned}$$

Bayes and expected value

Question 4 · A positive medical test

Let D denote having the disease, D' being healthy, and $+$ a positive test.

$P(D) = 0.02$, $P(D') = 0.98$, and $P(+ | D) = 0.95$.

The true negative rate is 0.90, so $P(+ | D') = 1 - 0.90 = 0.10$.

By total probability,

$$P(+) = P(+ | D)P(D) + P(+ | D')P(D')$$

$$= 0.95 \times 0.02 + 0.10 \times 0.98$$

$$= 0.019 + 0.098 = 0.117.$$

By Bayes' rule,

$$P(D | +) = P(+ | D)P(D) / P(+)$$

$$= 0.019 / 0.117 = 19/117 \approx \mathbf{0.162393 (16.2393\%)}$$

A frequency check: among 10,000 people, the model predicts 190 true positives and 980 false positives. Thus $190 / (190 + 980) = 19/117$. The low disease prevalence explains why a positive result does not imply a 95% chance of disease.

Question 5 · Mean, variance, and sample mean

x	85	90	95	100
$P(X = x)$	0.375	0.375	0.125	0.125

(a) Expected value

$$E[X] = \sum x P(X = x)$$

$$= 85(0.375) + 90(0.375) + 95(0.125) + 100(0.125)$$

$$= 31.875 + 33.75 + 11.875 + 12.5 = \mathbf{90}.$$

(b) Variance

$$E[X^2] = 85^2(0.375) + 90^2(0.375) + 95^2(0.125) + 100^2(0.125)$$

$$= 2709.375 + 3037.5 + 1128.125 + 1250 = 8125.$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 8125 - 90^2 = \mathbf{25}.$$

(c) Sample mean

For $\{85, 90, 85, 95, 90, 85, 100, 90\}$,

$$\text{sample mean} = (85 + 90 + 85 + 95 + 90 + 85 + 100 + 90) / 8$$

$= 720 / 8 = \mathbf{90}$, equal to $E[X]$. The sample proportions match the stated distribution exactly; this equality is not guaranteed for every random sample.

Information theory

Question 6 · Entropy in bits

The probabilities of messages A, B, C, and D are 0.4, 0.3, 0.2, and 0.1. Logarithms use base 2, so entropy is measured in bits.

(a) Original distribution

$$\begin{aligned} H(X) &= -\sum p(x) \log_2 p(x) \\ &= -[0.4 \log_2(0.4) + 0.3 \log_2(0.3) + 0.2 \log_2(0.2) + 0.1 \log_2(0.1)]. \end{aligned}$$

Message	p(x)	$-p(x) \log_2 p(x)$
A	0.4	0.52877124
B	0.3	0.52108968
C	0.2	0.46438562
D	0.1	0.33219281

Adding the contributions using unrounded values gives

$$H(X) = 1.84643934 \text{ bits} \approx 1.8464 \text{ bits}.$$

(b) Equal probabilities

With four equally likely messages, each probability is $1/4 = 0.25$.

$$H(X) = -4(0.25) \log_2(0.25) = -1(-2) = 2 \text{ bits}.$$

This is the maximum entropy for four possible messages: $\log_2(4) = 2$. Equal probabilities make the next message most uncertain.

Reproducibility and sources

Run `python theory/probability_solutions.py` from the project directory to reproduce all numerical answers. The script uses exact fractions where possible and checks variance by an independent formula.

Source: supplied Assignment1-1.pdf, Questions 1–6. Methods follow the supplied Module 2 probability practical. Prepared with OpenAI Codex assistance; review the reasoning and follow course disclosure requirements before submission.